

MATEO CASTANO-YEPES

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EDUCATION

University of Toronto | 3.7/4.0 cGPA

B.A.Sc. in Mechanical Engineering + PEY Co-op

Toronto, ON

Class of 2029 + PEY

WORK EXPERIENCE

Mechanical Engineer Intern (Part-Time)

San Diego, CA (Remote)

Space Copy

June 2026 – Present

- Aided in the development of autonomous ISRU **3D printing** systems for lunar missions under a phase II NASA contract.
- Spearheaded the adoption of autonomous heating, levelling and part-release mechanisms, modelled with **SolidWorks**.
- Researched different build plate material options with an emphasis on effectively 3D printing lunar regolith simulants.
- Fully rendered the CAD model of the 3D printer using **Blender** to accurately represent different materials and lighting.

Health & Safety Coordinator

Waterloo, ON

BrickWorks Academy

July 2026 – Present

- Implemented safety procedures, first aid and daily cleaning to maintain camper safety across 4 STEM summer camps.
- Instructed hundreds of campers on using **CAD** software, **3D printing**, **DFM**, and building/coding with LEGO Robotics.

Precast Manufacturing Engineer Intern

Melbourne, AUS

SH Engineering Pty Ltd

May 2026 – July 2026

- Led the integration of a hazardous gas alarm system with PLC infrastructure, modelled in **AutoCAD** and **SolidWorks**.
- Verified structural integrity and quality through inspections of 200+ precast segments for two Australian tunnel projects.
- Increased carousel production by **11%** by logging 5+ hours of production cycle times across the line (**Excel**), identifying main bottlenecks and producing a report with recommendations on how to distribute resources to decrease cycle times.
- Aided in the development of two 15+ page engineering proposals identifying various areas of improvement in the line to increase production by **33%** and increase segment storage capacity by **15,000+** to prevent overflow from production.
- Compiled daily shift report data into an organized production tracker, automating weekly average calculations in **Excel**.

PROJECT/EXTRACURRICULAR EXPERIENCE

Structural/Fabrication Team Member

Toronto, ON

UofT Blue Sky Solar Racing

July 2026 – Present

- Aiding in the development of a solar powered car for the American Solar Challenge, an 8-day 2,400+ km race.

Research Lead

Toronto, ON

TriggerPredict (For Mount Sinai Fertility Clinic)

January 2026 – April 2026

- Engineered **124** conceptual designs from a client statement, documented in three technical reports totaling 97 pages.
- Coordinated research objectives and tasks within a group of five on In-Vitro Fertilization trigger timing and ML models.
- Developed the TriggerPredict app end to end using **Python** and **Typescript**, scoring a **8.2/10** System Usability Score.

General Member

Toronto, ON

UofT Human Powered Vehicles Design Team

September 2025 – April 2026

- Developed CAD modelling proficiency in **SolidWorks** through technical workshops focused on aerodynamics.
- Fabricated composite material components for high-speed human-powered vehicles using hand layup techniques.

Build Team Member

Kitchener, ON

Rockway FIRST Robotics Team

September 2024 – May 2025

- Developed and tested robot mechanism prototypes using **3D printing** and woodworking to inform design decisions.
- Created full scale models of robot components in **Fusion**, maintaining accuracy through iterative design changes.

SKILLS

Technical: SolidWorks CSWA, Fusion, AutoCAD, Blender, ANSYS, Python, MATLAB, Excel, Machining, 3D Printing

Soft Skills: Prototyping/Problem Solving, Leadership, Project Management, Communication, Teamwork, Spanish